PREDICTING INTERNATIONAL ESCALATIONS AND NARRATIVE SHIFTS IN CONFLICT

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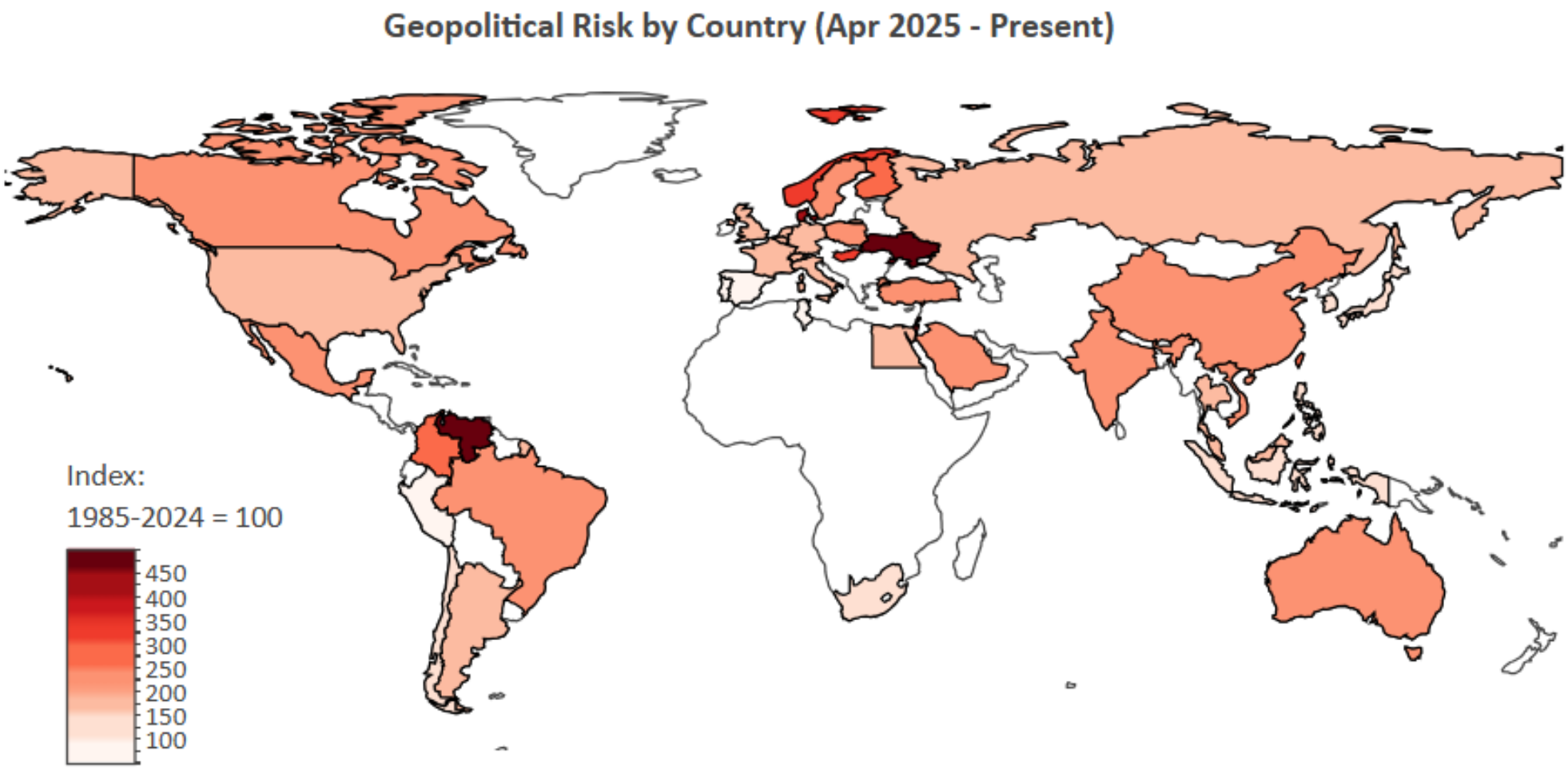


This project examines how conflict events and geopolitical risk signals drive escalation and how media narratives reflect these dynamics.

DATA SOURCES

1. ACLED Dataset (Armed Conflict Location & Event Data**)**

- Acquisition: CSV data exported from acleddata.com
- Conflict events processed : 1,554,214
- Date range: Jan 2015 → Mar 2025
- Countries: 52

2. GPR Index - Geopolitical Risk Index (goes up when more newspapers write about wars and crises.)

- Acquisition: XLS file downloaded from matteoiacoviello.com
- Daily file rows: 15,050 (1985 → 2026)
- Monthly file rows: 494
- After filtering from date(jan 2015- mar 2025) and week wise- 531 week rows
- Countries: 44

New York Times (393 articles)

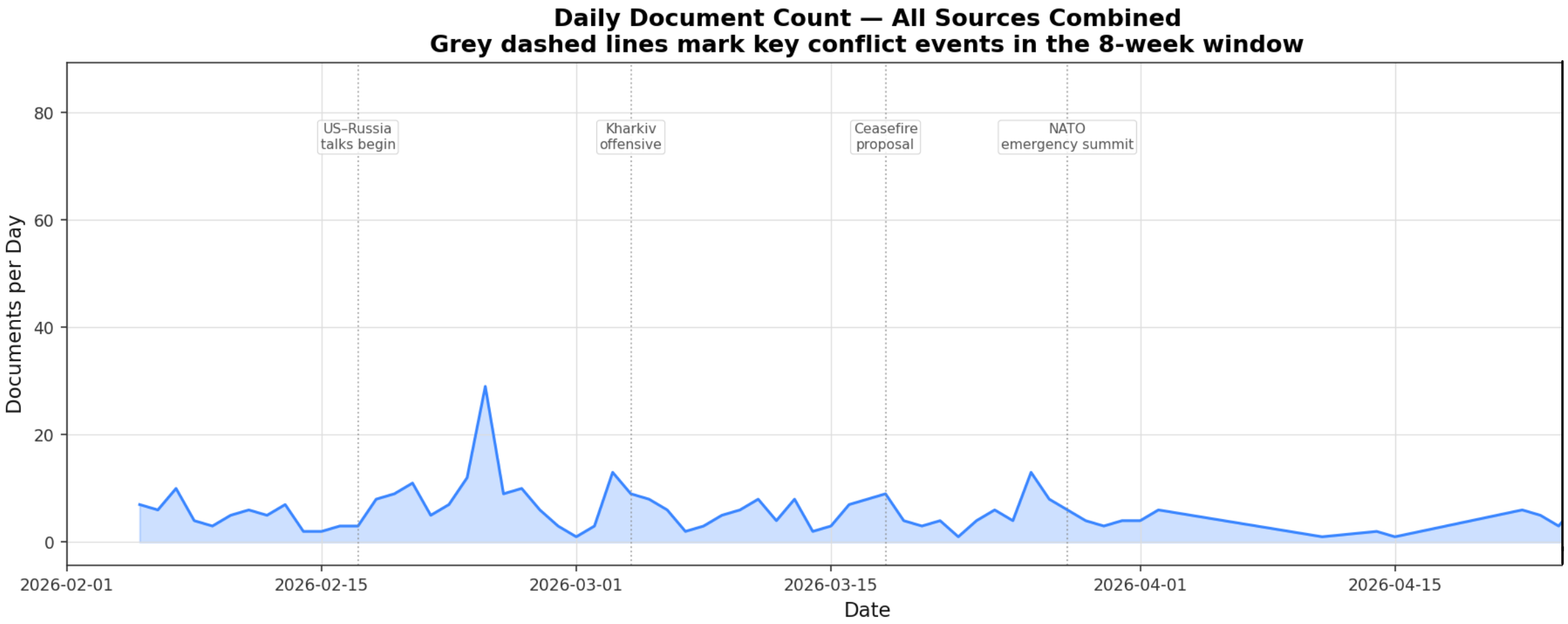
- API Text + Metadata
- Acquisition: developer.nytimes.com API key
- Fields: headline, abstract, pub_date, section
- Returns JSON files

Reddit (297 posts)

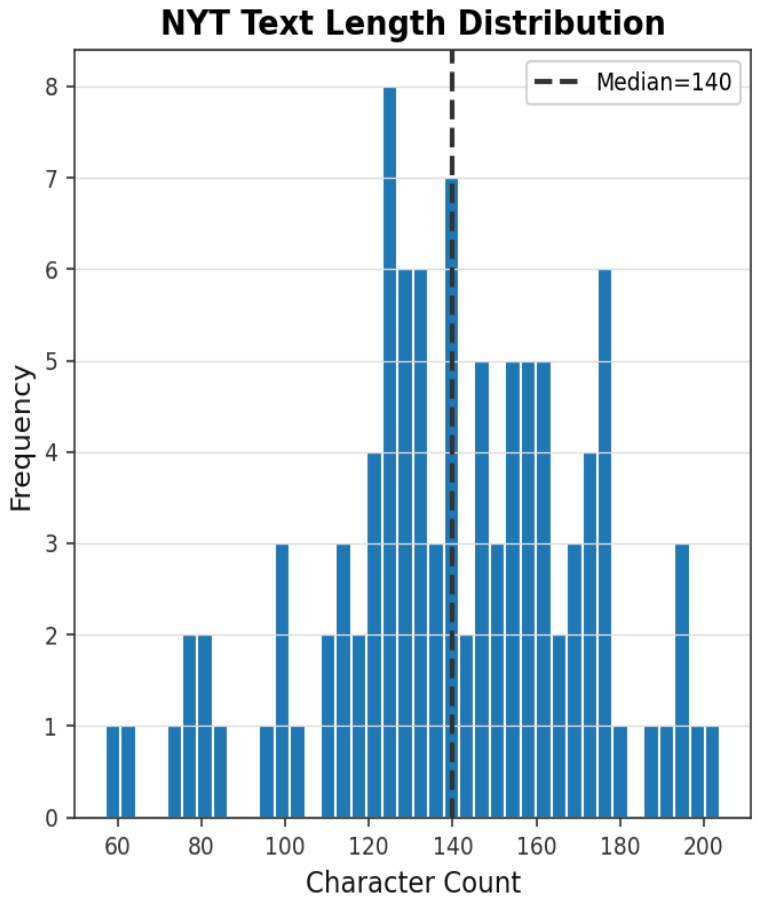
- Captured using PRAW API
- Subreddits: r/geopolitics, r/worldnews
- Fields: title, selftext, score (upvotes), num_comments, post_date

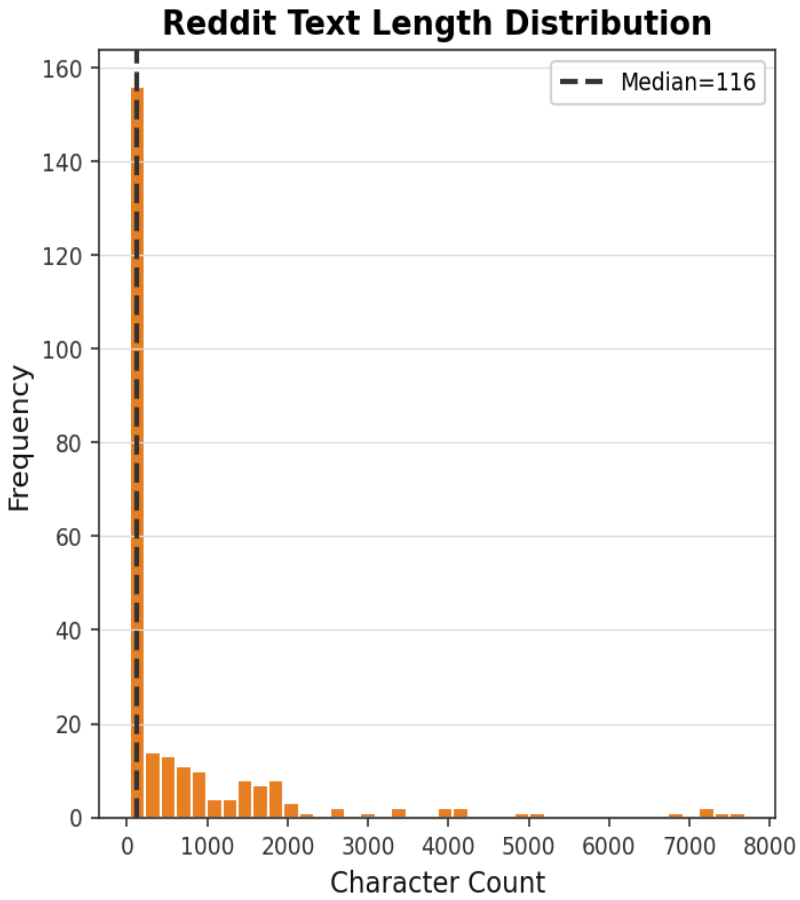
YouTube Data (287 uploads)

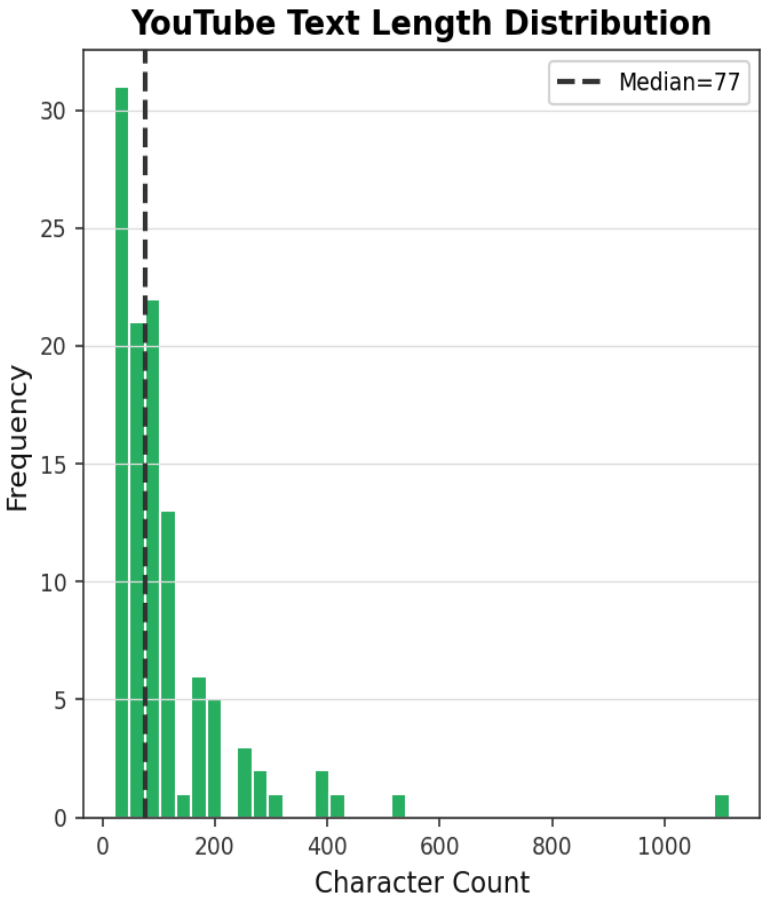
- Json from YouTube Data API
- Fields: Title, Description, Views, Likes
- collected from channels including Al Jazeera, DW, CGTN, and BBC



Media activity shows clear spikes during major conflict events







NYT

- Articles are **consistent and structured**

Reddit

- **Highly skewed distribution** with extreme long posts (up to ~7000 chars)

YouTube (Comments)

- Mostly **short, reactive, and opinion-based comments**
- Captures **real-time public sentiment and emotional reactions**

Data Preprocessing

- Text cleaning: removed URLs, HTML tags, and non-alphabetic characters
- Converted all text to lowercase
- Tokenization using NLTK
- Lemmatization applied

- Example:

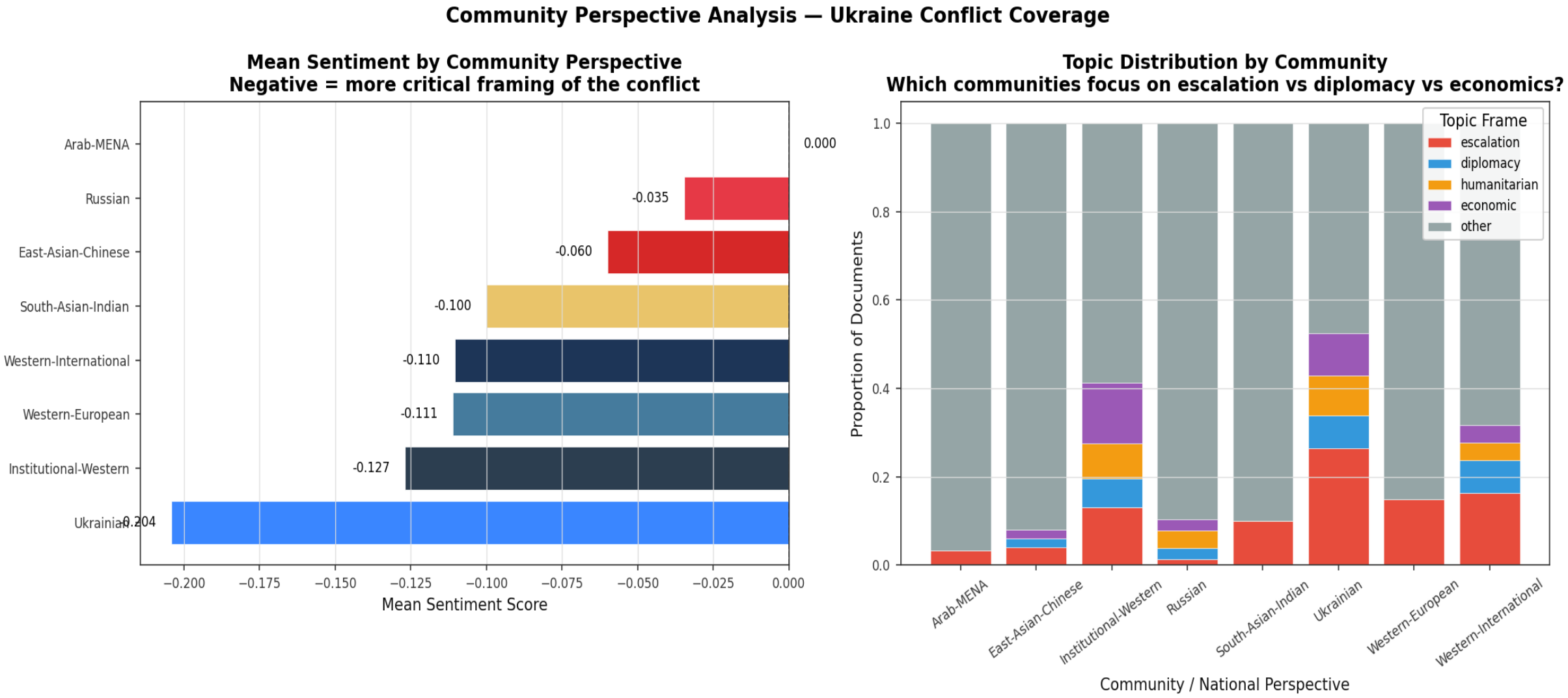
RAW: "war!!! http link"
CLEAN: "war"

lexicon-based topic classification method:

- **Escalation lexicon** → words related to conflict
- **Diplomacy lexicon** → words related to peace
- "escalation": ["attack","bomb","airstrike","invasion","war","troops","military"]
- "diplomacy": ["ceasefire","talks","agreement","peace","negotiation","truce"]

BERT(ro-BERT-a) model for sentiment analysis:

Transformer-based models (RoBERTa) capture context better than lexicon methods:
Example:
“airstrike launched after failed peace talks”
Match words against lexicons:
Negative sentiment - "airstrike"
Positive sentiment - "peace", "talks"



Negative sentiment reflects conflict-heavy framing, while variations across communities indicate differing geopolitical perspectives.

Feature Engineering:

- Aligned datasets on: Country, weekly
- Created unified dataset: gpr indices + Conflict metrics
- Created weekly features:

ACLED-derived weekly features

total_events: Count of events that week — numerical
total_fatalities: Sum of deaths — numerical integer
fatality_rate: Deaths per event — lethality signal
battle_fraction: Share of events that are battles
multi_country_spike: How many countries spiking at once

GPR-derived weekly features

gpr_mean: Weekly average GPR index
gpr_threat_mean: Threat-component average
gpr_act_mean:Actual-events component avg
threat_act_ratio: Threat ÷ Act — pre-escalation signal

NLP media features (from APIs)

narrative_shift_score: Escalation topic% − Diplomacy topic%
mean_sentiment: Avg sentiment [−1, +1]

formula:

Augmented GPR = GPR (most recent) + Media Signal

Where:

Media Signal = Narrative Shift + Negative Sentiment

final version:

Augmented GPR = GPR + (Narrative Shift + Negative Sentiment)

Escalation Label Design

- **ESCALATION = 1 IF:**

  (FATALITIES IN 30 DAYS > 2 × AVERAGE FATALITIES OVER PAST 12 WEEKS)
  AND
  (TENSION IN 30 DAYS > 2 × AVERAGE TENSION OVER PAST 30 DAYS)

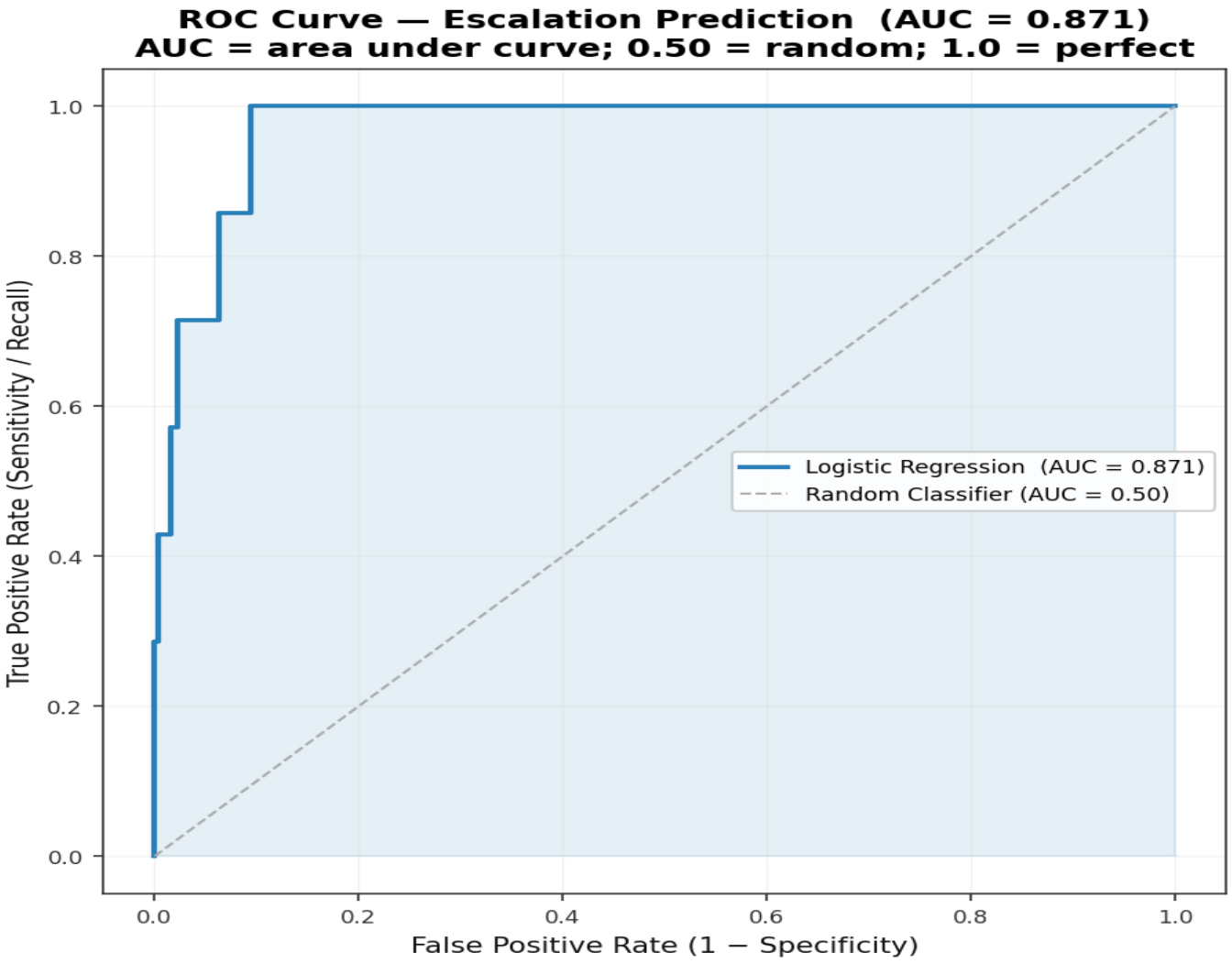
- **0 – STABLE**

- **1 - ESCALATION**

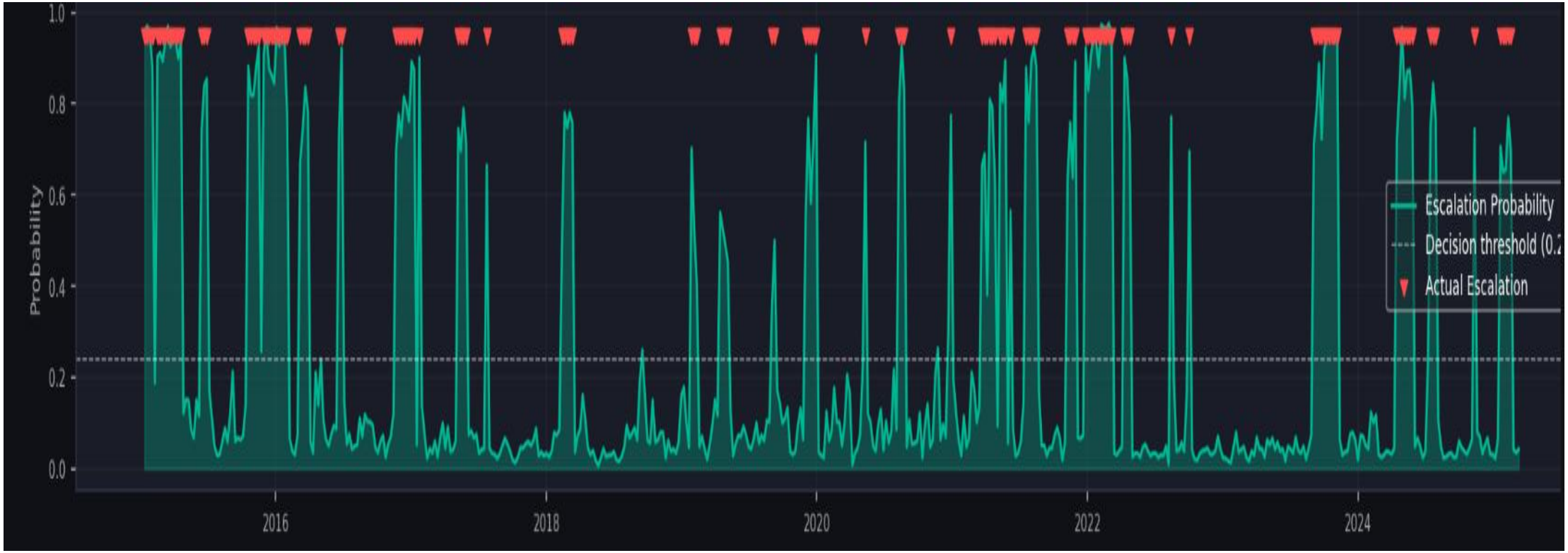
Version	Criteria	T-value	positives	remarks
1.a	OR	1.5	297/531	Too loose (HIGH FALSE POSITIVES)
1.b	AND	1.5	210/531	Loose
2.a	OR	3.0	62/531	Strict (MISSES EVENTS)
2.B	AND	3.0	7/531	Too strict
3.a	OR	2.0	182/531	Moderate
3.B	AND	2.0	130/531	Optimal

OUTPUT

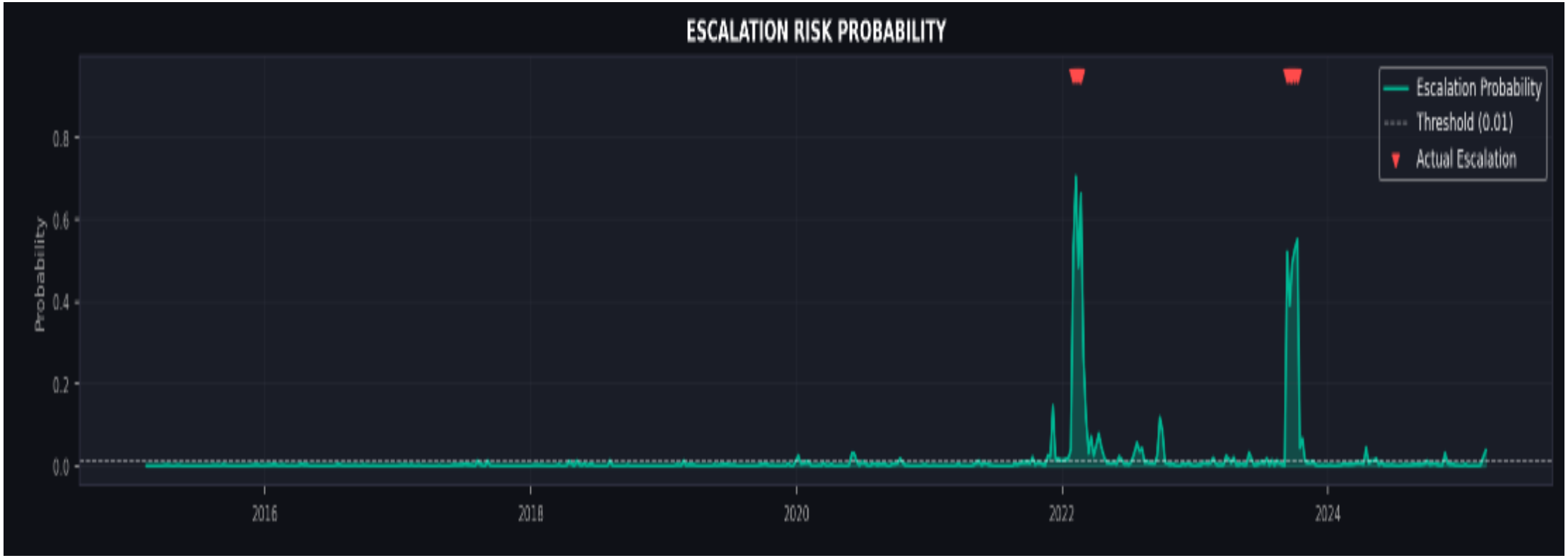
model	AUC
Logistic Regression	**0.871**
Gradient Boosting	0.541
XG Boost	0.693
Random Forest	0.57



Escalation Risk Probability [version 3.b]- optimal



Failed version[2.b]



Thank you